



12. A CS Stage is similar to which BJT Amplifier
 A) CC Stage B) Emitter Follower
 C) CE Stage D) CB Stage
13. To use MOSFET as an amplifying device we should operate it in
 A) Cut-off B) Saturation
 C) Triode D) Deep Triode
14. The zero depth of the channel at the drain end in the drain characteristics gives rise to the term
 A) junction capacitance B) transconductance
 C) channel pinch-off D) oxidation
15. The operation of a MOSFET in one of the regions resembles that of
 A) intrinsic semiconductor device B) vacuum-tube device
 C) compound semiconductor device D) carbon nano tube device
16. Which of the following is true about Source Follower?
 A) It has low input impedance B) It has high output impedance
 C) Infinite gain D) Output will be same as input
17. The maximum efficiency of a Class – B amplifier is
 A) 25% B) 55%
 C) 78% D) 87%
18. A centre-tapped transformer is used in
 A) Class – A B) Class – AB
 C) Class – B push pull D) Class – C with single supply
19. How many transistors must be used in a class B power amplifier to obtain the output for the full cycle of the signal?
 A) 2 B) 1
 C) 3 D) 4
20. A Differential Amplifier amplifies
 A) Input signal with higher voltage B) Input voltage with smaller voltage
 C) Sum of the input voltage D) Difference of the input voltage

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- 1 a Draw the circuit of an emitter bias configuration indicating the necessary parameters in it. Derive expressions for I_B , I_C and V_{CE} .
 b) Given the load line of Fig. 1.b) and the defined Q - point, determine the configuration required values of V_{CC} , R_C , and R_B for a fixed-bias configuration. Design the final circuit.

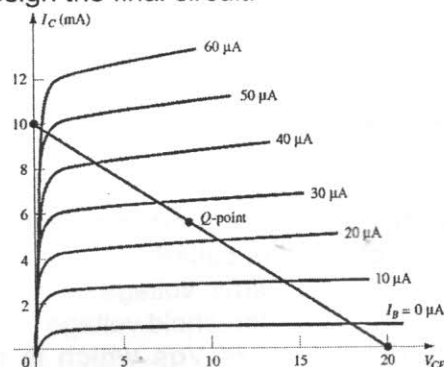


Figure. 1.(b)

2. a) Explain a voltage divider biasing configuration with a neat circuit diagram and parameters. Derive expressions for I_B and V_{CE} .

Marks	BT*	CO*	PO*
8	L*2	1	1

8	L3	1	2
8	L2	1	1

- b) Determine the dc level of I_B and V_C for the network of Fig. 2.b).

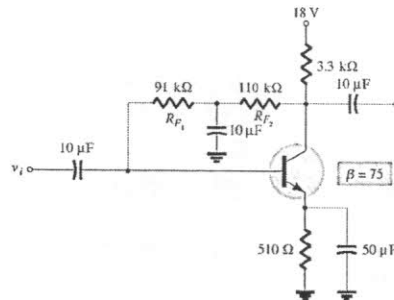


Figure. 2(b)

3. a) Explain early voltage and r_e model for BJT. Deduce the equivalent model of a BJT.
b) Write the r_e model for a circuit of fixed bias configuration. Derive input impedance, output impedance and voltage gain.

Unit – II

4. a) Explain the various parameters of drain current and their importance of the same with respect to MOSFET.
b) Consider a process technology for which $L_{\min} = 0.4 \mu\text{m}$, $t_{\text{ox}} = 8 \text{ nm}$, $\mu_n = 450 \text{ cm}^2/\text{V}^2$, and $V_t = 0.7 \text{ V}$.
(a) Find C_{ox} and k_n'
(b) For a MOSFET with, calculate the values of V_{OV} , V_{GS} , and V_{DSmin} needed to operate the transistor in the saturation region with a dc current $I_D = 100 \mu\text{A}$.
(c) For the device in (b), find the values of V_{OV} and V_{GS} required to cause the device to operate as a 1000Ω resistor for very small v_{DS} . Assume $\epsilon_{\text{ox}} = 3.45 \times 10^{-11} \text{ SI units}$.

5. a) Write the large-scale equivalent circuit model of n-channel MOSFET operating in the saturation condition, incorporating the output resistance.
b) Design the circuit in Fig. 5.b) to establish a drain voltage of 0.1 V .

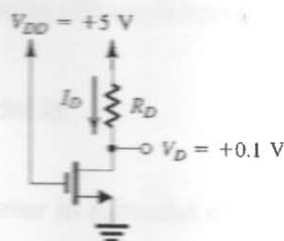


Figure. 5.(b)

What is the effective resistance between drain and source at this operating point? Let $V_{\text{tn}} = 1 \text{ V}$, $k_n'(W/L) = 1 \text{ mA/V}^2$.

6. a) With neat circuit diagram and hybrid-pi model, explain a Common Source amplifier.
b) Deduce an expression for voltage gain for the MOSFET working as a small signal amplifier.

Unit – III

7. a) Explain the concept of current mirror. With the necessary circuits and equations, explain a NMOS current mirror.
b) Discuss the MOS-differential pair with small signal analysis.

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|-------|---|---|----|---|---|
| 8. a) | Discuss the principle of operation of Class-B push pull amplifier with the necessary diagrams. | 8 | L2 | 5 | 1 |
| b) | A class A transformer coupled audio amplifier is required to deliver a maximum power of 1Watt to a loud speaker of 10Ω resistance. If the output resistance of amplifier is 1000Ω , Calculate | | | | |
| | (i) the turns ratio of the transformer required | | | | |
| | (ii) power supply voltage assuming ideal transformer. | 8 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
